Title

Development of a Digital Mental Health and Psychological Support System for Students in Higher Education.

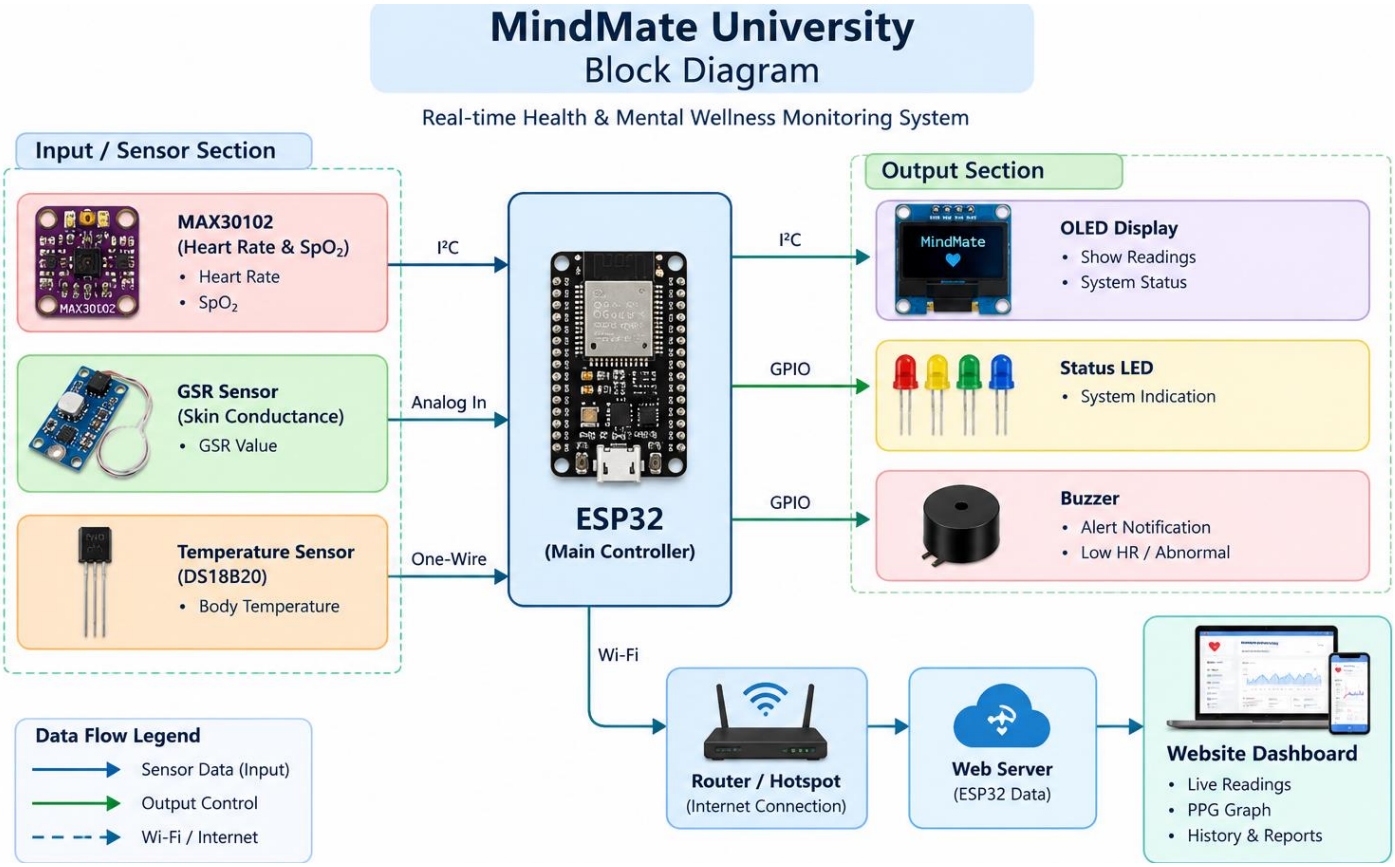
Methodology

Mental wellness is an important aspect of overall health, particularly among students experiencing academic pressure and emotional stress. MindSense is a proposed low-cost IoT-based system that combines physiological monitoring with a web-based mental wellness support platform.

The ESP32 microcontroller serves as the main processing and wireless communication unit. The MAX30102 sensor collects photoplethysmography (PPG) signals to estimate heart rate and peripheral oxygen saturation (SpO_2). A Galvanic Skin Response (GSR) sensor measures changes in skin conductance, which may indicate physiological arousal. A temperature sensor records temperature readings. An OLED display provides local feedback, while LEDs and a buzzer provide notifications under configured conditions.

The ESP32 transmits available sensor readings through Wi-Fi to the MindMate University web dashboard. The website presents heart rate, SpO_2, GSR, temperature, and a live PPG waveform. It also includes mental wellness questionnaires, assessment results, privacy information, counselling resources, and emergency contact information. Testing will evaluate sensor connectivity, data transmission, dashboard updates, and alert operation. Missing or invalid readings will be shown as unavailable rather than replaced with artificial values.

Proposed Block Diagram



Proposed Outcome

The expected outcome is a functional prototype demonstrating real-time physiological data monitoring and web-based visualization. The system aims to improve awareness of physiological changes and provide convenient access to mental wellness resources. It is designed for educational and wellness-awareness purposes only; sensor readings and questionnaire results cannot independently diagnose stress, anxiety, or other mental health conditions.